Fiscal Year:	FY 2026	**Task Last Updated:**	FY 09/16/2025
PI Name:	Paletz, Susannah Ph.D.		
Project Title:	Enhancing Performance and Communication for Distributed Teams During Lunar Spacewalks		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **HSIA**:Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture (2) **Team**:Risk of Performance and Behavioral Health Decrements Due to Inadequate Cooperation, Coordination, Communication, and Psychosocial Adaptation within a Team		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	paletz@umd.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	301-405-1218
Organization Name:	University of Maryland		
PI Address 1:	College of Information		
PI Address 2:	2117D Hornbake Building, South Wing		
PI Web Page:			
City:	College Park	**State:**	MD
Zip Code:	20742-4345	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2024 HERO 80JSC024NA001-FLAGSHIP, OMNIBUS, and NSCOR Crew Health: Topics in Human Health Countermeasures, Behavioral Performance, and Space Radiation and NASA Human Research Program Omnibus Opportunity
Start Date:	11/13/2024	**End Date:**	09/30/2025
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2025 per A. Beitman/HFBP (Ed., 8/14/25). NOTE: Period of performance (POP) has changed per NASA JSC and NSSC documentation. The new POP is 11/13/2024-11/12/2029. The original POP was: 10/01/2024-09/30/2029 (Ed., 3/31/2025).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bell, Suzanne Ph.D. (NASA Johnson Space Center) Landon, Lauren Ph.D. (KBR/NASA Johnson Space Center) Miller, Matthew Ph.D. (Jacobs Technology, Inc.) Mitra, Paromita M.S. (NASA Johnson Space Center) Karasinski, John Ph.D. (NASA Ames Research Center) Marquez, Jessica (NASA Johnson Space Center)		
Grant/Contract No.:	80NSSC25K7248		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	NASA's current human systems integration architecture relies on mission control teams to supervise and direct astronauts during spacewalks, and relies on continuous, real-time video feeds to maintain crew safety and operational success. Artemis extravehicular activities (EVAs) present new challenges, including communication delays, increased cognitive demands, and an increasingly distributed team. This proposal characterizes and mitigates risks associated with upcoming Artemis missions by developing and validating communication countermeasures to effectively enhance multi-team EVA performance. We will evaluate the impact of communication delay, information management, and shifting roles on team process and performance within a distributed multi-team EVA in the NASA Human Exploration Research Analog (HERA). We will develop and validate countermeasures to enable effective communication, facilitate EVA resource management, and enhance overall team performance.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	EVAs are one of the most safety-critical spaceflight tasks, often tied to mission-level goals and objectives of lunar surface missions. Unfortunately, the multi-team processes and performance remain largely unquantified, which is particularly risky as we enter the Artemis era of lunar exploration. Our proposed research is novel and unique, as it will be among the first to investigate multi-team performance in lunar EVAs under Artemis-like operational conditions – including a simulated lunar environment, communication delays, and a modern digital EVA information system. As a result, we will identify, evaluate, and validate communication countermeasures that support Artemis EVAs and facilitate future EVA resource management.
Task Progress:	Unlike International Space Station (ISS) extravehicular activities (EVAs), future Artemis spacewalks will challenge astronauts to conduct novel tasks as they explore and traverse unknown lunar terrain. Compared to ISS and Apollo missions, Artemis EVAs will face new challenges. These include modern digital information technologies, communication transmission delays, high physical workload, and increased EVA frequency with associated physiological challenges, which include increased cognitive demands, information overload, and the need to coordinate larger, more distributed teams. NASA is addressing this EVA evolution through its Moon-to-Mars architecture, emphasizing maximizing crew time for high-value activities. However, the impact of these factors on multi-team performance during lunar EVAs, one of the most safety-critical spaceflight tasks and primary objectives for lunar surface missions, remains largely unquantified. This research aims to identify communication countermeasures that best support multi-team performance and facilitate resource management for lunar surface EVA operations. Aim 1: Evaluate the impact of lunar communication delay, information management, and novel roles on team processes and performance within a distributed multi-team EVA. Aim 2: Develop, test, and validate countermeasures to enable effective communication, facilitate EVA resource management, and enhance overall team performance. During Year 1, the focus was on defining an experiment to be completed in an analog simulating lunar exploration. Aside from preparing, submitting, and receiving approval from NASA's Institutional Review Board, we established a reliance agreement with the University of Maryland. Our team defined study objectives, primary measures, and data collection plan. Two EVA procedures were defined as well. Procedures are the step-by-step instructions given to the two analog astronauts tracking the simulated EVA and executed by the two other analog astronauts in the virtual reality EVA system. EVA procedures are being formatted appropriately to follow the expected organization of ISS EVA procedures. During Year 1, we presented at the annual INGRoup conference, examining how information flows across a multiteam system to inform future countermeasure design. Effective teamwork depends on communication that is timely, intuitive, and relevant to the task at hand. Communication delays and non-colocation harms mental model sharedness and accuracy, impairs situational awareness, increases cognitive load, and hinders information flow and team coordination. Well-designed communication tools and protocols, paired with strategically distributed roles and responsibilities, may improve shared mental models and EVA performance. Additionally, we conducted a second comprehensive literature review designed to systematically analyze the transformative evolution of spacesuit advanced informatics, display, and command and control designs, concurrently examining the technological implications for the supporting ground infrastructure. The methodology for this review involves an identification and synthesis of publicly available scientific and technical literature compiled over the past four decades. An abstract was submitted and accepted for this review. The paper will be published in IEEE Aerospace Conference in the winter of 2026.
Bibliography Type:	Description: (Last Updated: 10/01/2025)
Abstracts for Journals and Proceedings	Landon LB, Paletz SBF, Karasinski JA, Miller MJ, Keller JR, Bell ST, Marquez JJ. "Leveraging distributed teams and human-computer interaction to coordinate in adverse conditions." 20th Annual INGRoup conference, Rotterdam, Netherland, July 17-19, 2025. Abstracts. 20th Annual INGRoup conference, Rotterdam, Netherland, July 17-19, 2025. , Jul-2025